FEDERAL SPECIFICATION

NEEDLE, HYPODERMIC, BLOOD COLLECTING, PETROFF

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers hypodermic needle, Petroff design, used in the collection of blood.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

DD–T–736—Tubes; Test, Laboratory, Glass.

GG–I–526—Instruments, Dental and Surgical; General Specifications.

GG–N–196—Needles, Hypodermic; for Luer Syringes.

PPP–B–566—Boxes, Folding, Paperboard.

PPP–B–585—Boxes, Wood, Wirebound.

PPP–B–601—Boxes, Wood, Cleated-Plywood.

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Box, Fiberboard.

PPP–B–676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration, Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–117—Bags, Interior Packaging.

MIL–L–10547—Liners, Case, Waterproof.

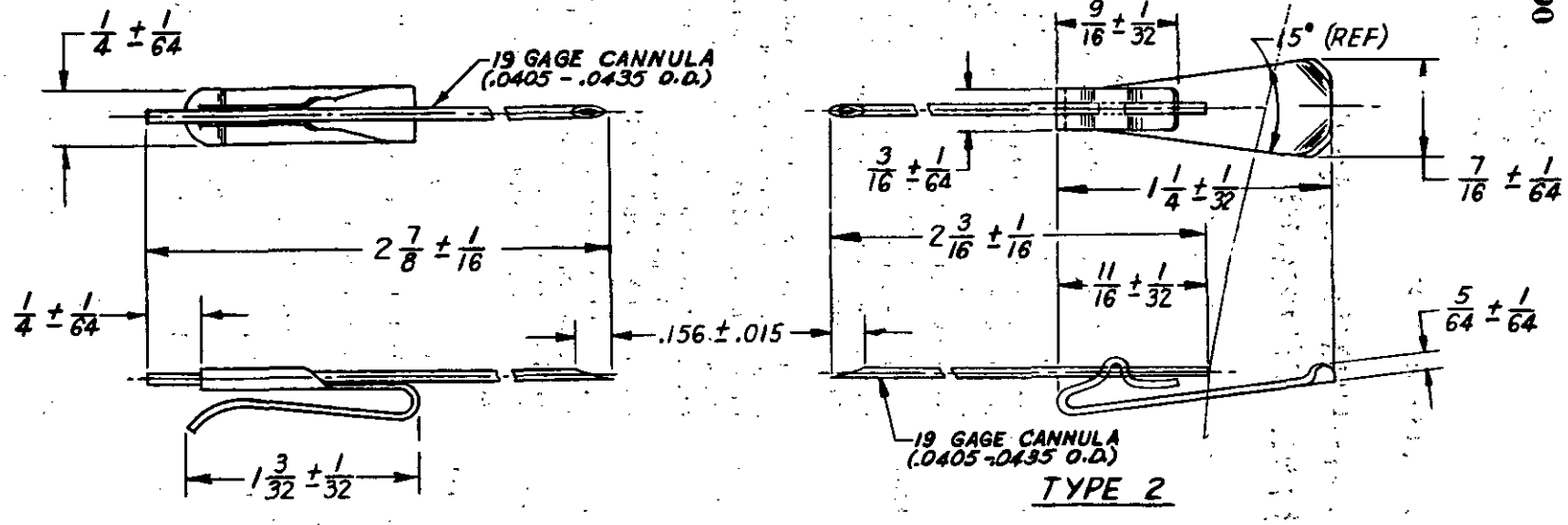
Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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NOTE

1. STYLETTE SHALL BE PRACTICAL FOR THE CANNULA I.D. AND NOT LESS THAN 1/4 LONGER THAN OVERALL LENGTH OF NEEDLE IS TO BE INSERTED IN NEEDLE.
2. THICKNESS OF CLIP = .023 ± .003.
3. CANNULA I.D. = .0255 MIN.

FIG. 1

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3. REQUIREMENTS

3.1 Material. Needle shall be fabricated from materials in accordance with Federal Specification GG–I–526, as follows:

Component	Specification designation	
	Class	Composition
Cannula	–	AISI 304[1]
Spring clip	4	N[2]
Wire	2	G or (nickel wire)

[1] Shall comply with the boil test requirements of class 2, composition G material of Federal Specification GG–I–526.

[2] Except shall be nickel plated only.

3.2 Style, design, and dimensions. Blood collecting hypodermic needle shall be of the Petroff design, in conformance with type I or type II of figure 1. A spring clip shall be rigidly and permanently attached to the cannula and shall be so constructed that it will firmly attach the needle to the lip of a 13 by 100 millimeter (mm.) Wasserman test tube (see 3.5).

3.3 Stiffness, elasticity, and resistance to breakage. Stiffness, elasticity, and resistance to breakage shall conform to table II, for 19 gage, type I needle, of Federal Specification GG–N–196.

3.4 Security of assembly. The juncture of the clip and cannula shall not separate when the needle is subjected to the test specified in 4.4.1.

3.5 Spring clip tension. When tested as specified in 4.4.2, the test tube shall not become detached from the spring clip.

3.6 Additional requirements. Except as specified herein, the needle shall comply with the applicable requirements and tests of Federal Specification GG–I–526. The requirements of that specification for micrometallographic examination shall not apply for class 2 material.

3.7 Workmanship. Workmanship shall be first class throughout. Needles shall be free from defects which detract from their appearance or may impair their serviceability. In addition, the lumen of the cannula shall show complete freedom from metal chips and other foreign matter.

3.8 Finish. Needle shall have a smooth bright finish.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 *Certificates of quality.* When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling for lot acceptance.

4.2.1 *Inspection lot.* The inspection lot shall be determined by the cognizant government inspector and shall be based on the unit of product being one needle (end item).

4.2.2 *Sampling for examination.* A random sample of needles shall be selected from each lot offered to the government in accordance with

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Military Standard MIL–STD–105. Examination for dimensions shall be in accordance with Military Standard MIL–STD–105, and as indicated in table I.

4.2.3 *For tests.* Sampling for tests shall be conducted in accordance with Military Standard MIL–STD–105, and as indicated in table I.

4.2.4 *Raw materials.* A sufficient quantity of each lot of each material used in the construction of the needles shall be selected for chemical analysis.

4.2.5 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The needles shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 *Security of assembly.* With the clip in a fixed position, a 15-pound weight shall be suspended from the cannula for a period of 1-minute.

4.4.2 *Spring clip tension test.* The blood collecting needle shall be secured at the cannula, close to the pointed end, in a vertical position with the spring clip at the lower position. A test tube, filled with water, shall then be suspended from the spring clip. The test tube shall be a 13 by 100 mm. Wasserman test tube in accordance with type II, class B of Federal Specification DD–T–736.

5. PREPARATION FOR DELIVERY

For civil agency procurement only, Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging and packing protection for supplies and equipment.

5.1 Packaging.

5.1.1 *Level A.* Each needle shall be unit packaged in an envelope or carton of suitable size and design. Envelope or carton shall be adequately secured to prevent accidental opening. The completed package shall be clean and free of dirt, dust, or other foreign matter, and shall maintain the needle in the degree of cleanliness specified in 3.7.

TABLE I.—*Sampling criteria*

Component	Characteristic	Req.	Test procedure	Unit of product	Insp. level	AQL (percent defective)
Needle (end item)	Stiffness test	3.3	GG–N–196	1—needle	L2*	**
	Elasticity test	3.3	GG–N–196	1—needle	L2*	**
	Resistance to breakage	3.3	GG–N–196	1—needle	L2*	**
	Security of assembly	3.4	4.4.1	1—needle	L3*	**
	Spring clip tension	3.5	4.4.2	1—needle	L3*	**
	Dimensions	3.2	.....	1—needle	L3	**
Cannula	Boil test	3.1	GG–I–526	1—needle	L3*	**
Spring clip	Boil test	3.1	GG–I–526	1—needle	L3*	**
	Salt spray test	3.1	GG–I–526	1—needle	L3*	4.0

* A minimum of three units shall be tested regardless of lot size.
** Acceptance number shall be zero for all sample sizes.

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5.1.1.1 *Quantities.* Needles shall be packaged in intermediate, protective, and exterior containers, in the quantities specified in table II.

TABLE II.—*Packaging quantities*

Intermediate package	Protective package	Exterior container
12 needles	144 needles	864 needles

5.1.1.2 *Packing variation permitted.* If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.1.3 *Intermediate package.* Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening. Each carton shall be further packaged in a bag conforming to Military Specification MIL–B–117, type II, class d. Bag shall be fabricated from a combination of wet-strength kraft and metal foil.

5.1.1.4 *Protective package.* Protective package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, type I, class 1, or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.2 *Level B.* Each needle shall be packaged as specified in 5.1.1 and intermediately packaged as specified in 5.1.1.3, except barrier bag will not be required.

5.1.2.1 *Protective package.* Protective package shall be as specified in 5.1.1.4.

5.1.3 *Level C.* Needles shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.* Needles packaged as specified in 5.1.1 shall be packed in shipping containers designed for type 2 load, constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; or PPP–B–621, table III. Each box shall be lined with a waterproof case liner, conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.* Needles packaged as specified in 5.1.2 shall be packed in shipping containers constructed in accordance with Federal Specification PPP–B–621, table I, or PPP–B–636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

5.2.3 *Level C.* The subject commodity shall be packed in substantial commercial containers of the type, size, and kind, commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.*

5.3.1.1 *Shipping containers.* In addition to any special marking specified in the contract or order, containers shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Unit package.* Each unit package shall be marked with stock number, item identification, unit, quantity, and name of contractor.

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5.3.2.2 *Intermediate package.* Each intermediate carton and bag, when used, shall be marked in accordance with Military Standard MIL–STD–129.

5.3.2.3 *Protective package.* Protective package shall be marked in accordance with Military Standard MIL–STD–129.

5.3.2.4 *Exterior container.* Exterior container shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(1) Level B preservation and packaging is intended to provide economical but limited protection, and should be specified only when it is determined the needles will be held in covered storage approximately 12 months from date of initial packaging.

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Needles, not otherwise indexed by name
Carload minimum weight 20,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Needles, not otherwise indexed.
Motor volume minimum weight 20,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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